

EX No:1	EDA-DATA CLEANING

Aim:

To perform Exploratory Data Analysis (EDA) and data cleaning on the City Day Air Quality dataset using Pandas, NumPy, and Matplotlib by handling missing values, removing duplicates, converting data types, and analyzing air quality patterns.

Algorithm:

1. Import pandas, numpy, and matplotlib.pyplot.
2. Load city_day.csv and inspect the dataset structure, missing values, and duplicates.
3. Remove duplicate records and handle missing values in the dataset.
4. Convert the Date column into datetime format.
5. Perform EDA on AQI, AQI_Bucket, City, and pollutant columns.
6. Plot frequency distributions and a correlation heatmap.
7. Summarize the air quality insights and end execution.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# =====
# CRISP-DM FRAMEWORK
# =====

# 1. BUSINESS UNDERSTANDING

# Objective:
# Analyze air quality patterns, AQI levels,
# pollutant concentrations and city-wise trends.

# 2. DATA UNDERSTANDING

df = pd.read_csv("city_day.csv")
print(df.head())
print(df.shape)
print(df.info())
print(df.describe())
```

```

print(df.isnull().sum())
print(df.duplicated().sum())
# 3. DATA PREPARATION
df.drop_duplicates(inplace=True)
df['Date'] = pd.to_datetime(df['te'])
# Fill numerical missing values using median
num_cols = df.select_dtypes(include=np.number).columns
df[num_cols] = df[num_cols].fillna(df[num_cols].median())
# Fill categorical missing values using mode
cat_cols = df.select_dtypes(include='object').columns
for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
# 4. MODELING (EDA)
df['City'].value_counts().head(10).plot(kind='bar')
plt.title("Top Cities in Dataset")
plt.show()
df['AQI_Bucket'].value_counts().plot(kind='bar')
plt.title("AQI Category Distribution")
plt.show()
df['AQI'].plot(kind='hist', bins=30)
plt.title("AQI Distribution")
plt.xlabel("AQI")
plt.show()
df['PM2.5'].plot(kind='hist', bins=30)
plt.title("PM2.5 Distribution")
plt.xlabel("PM2.5")
plt.show()
corr = df.select_dtypes(include='number').corr()
plt.figure(figsize=(10,8))
plt.imshow(corr)
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)

```

```
plt.xticks(range(len(corr.columns)), corr.columns)
plt.title("Correlation Heatmap")
plt.show()
# 5. EVALUATION
print("Air quality patterns and trends identified successfully.")
# 6. DEPLOYMENT
print("Insights can be deployed through dashboards.")
# =====
# SEMMA FRAMEWORK
# =====
# SAMPLE
df = pd.read_csv("city_day.csv")
# EXPLORE
print(df.head())
print(df.info())
print(df.describe())
print(df.isnull().sum())
# MODIFY
df.drop_duplicates(inplace=True)
df['Date'] = pd.to_datetime(df['Date'])
num_cols = df.select_dtypes(include=np.number).columns
df[num_cols] = df[num_cols].fillna(df[num_cols].median())
cat_cols = df.select_dtypes(include='object').columns
for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
# MODEL
df['AQI_Bucket'].value_counts().plot(kind='bar')
plt.title("AQI Category Distribution")
plt.show()
df['AQI'].plot(kind='hist', bins=30)
plt.title("AQI Distribution")
plt.show()
```

```
df['City'].value_counts().head(10).plot(kind='bar')
plt.title("Top Cities")
plt.show()
corr = df.select_dtypes(include='number').corr()
plt.figure(figsize=(10,8))
plt.imshow(corr)
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.title("Correlation Heatmap")
plt.show()
# ASSESS
print("Patterns and trends successfully identified.")
# =====
# KDD FRAMEWORK
# =====
# SELECTION
df = pd.read_csv("city_day.csv")
# PREPROCESSING
print(df.info())
print(df.isnull().sum())
print(df.duplicated().sum())
df.drop_duplicates(inplace=True)
# TRANSFORMATION
df['Date'] = pd.to_datetime(df['Date'])
num_cols = df.select_dtypes(include=np.number).columns
df[num_cols] = df[num_cols].fillna(df[num_cols].median())
cat_cols = df.select_dtypes(include='object').columns
for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
# DATA MINING
df['AQI_Bucket'].value_counts().plot(kind='bar')
```

```

plt.title("AQI Category Distribution")

plt.show()

df['AQI'].plot(kind='hist', bins=30)

plt.title("AQI Distribution")

plt.show()

df['City'].value_counts().head(10).plot(kind='bar')

plt.title("Top Cities")

plt.show()

corr = df.select_dtypes(include='number').corr()

plt.figure(figsize=(10,8))

plt.imshow(corr)

plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)

plt.yticks(range(len(corr.columns)), corr.columns)

plt.title("Correlation Heatmap")

plt.show()

# INTERPRETATION

print("Knowledge discovered from the dataset has been interpreted to support air quality analysis.")

```

Output:

```

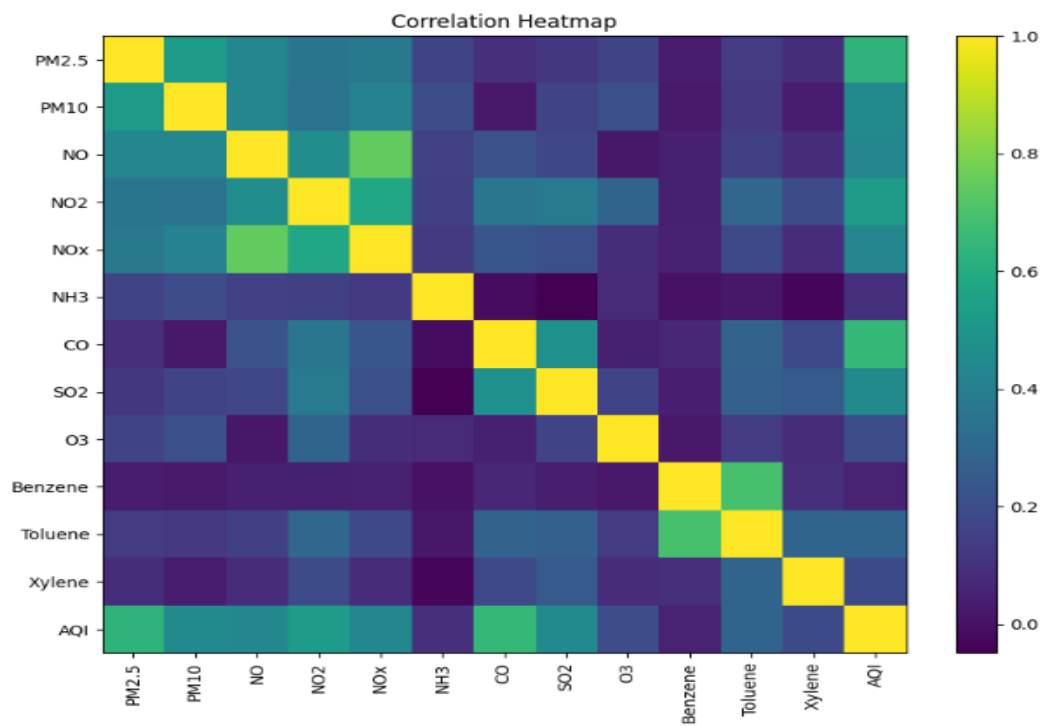
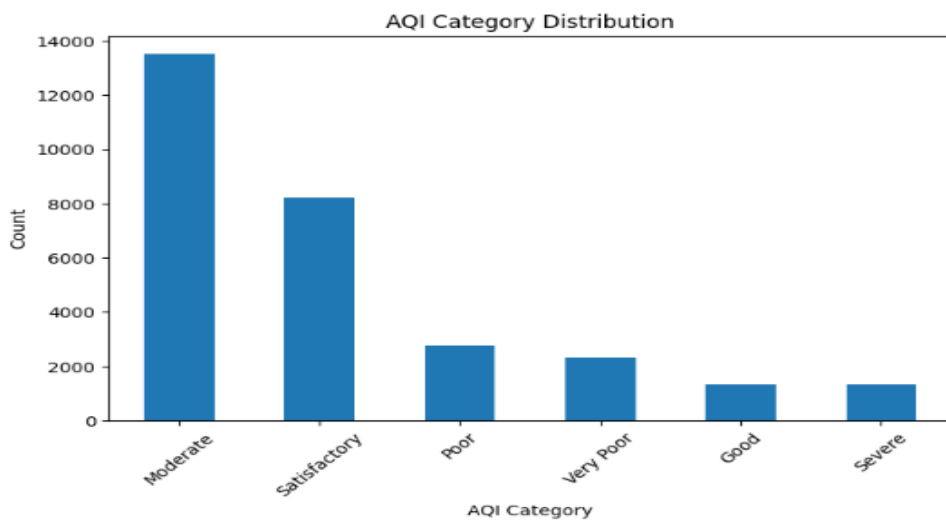
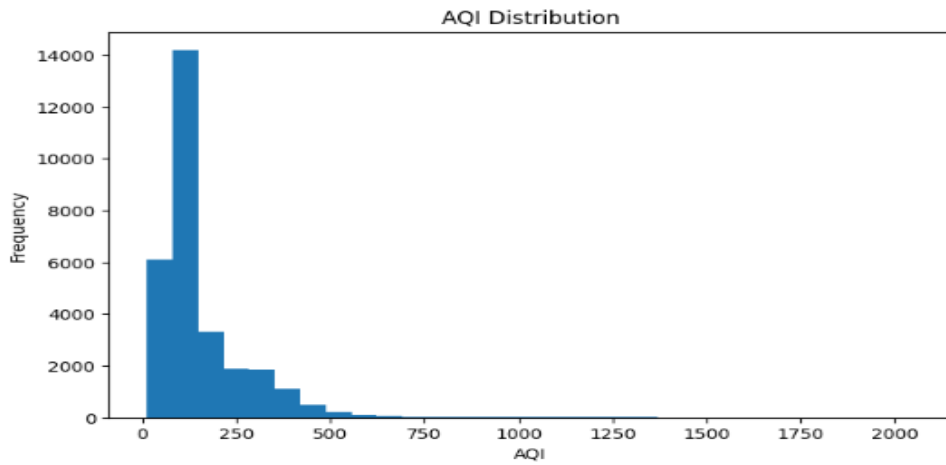
First 5 Records:
   City      Date  PM2.5  PM10    NO    NO2    NOx   NH3    CO    SO2
0  Ahmedabad  2015-01-01   NaN   NaN   0.92  18.22  17.15   NaN   0.92  27.64
1  Ahmedabad  2015-01-02   NaN   NaN   0.97  15.69  16.46   NaN   0.97  24.55
2  Ahmedabad  2015-01-03   NaN   NaN  17.40  19.30  29.70   NaN  17.40  29.07
3  Ahmedabad  2015-01-04   NaN   NaN   1.70  18.48  17.97   NaN   1.70  18.59
4  Ahmedabad  2015-01-05   NaN   NaN  22.10  21.42  37.76   NaN  22.10  39.33

   O3  Benzene  Toluene  Xylene  AQI  AQI_Bucket
0  133.36    0.00    0.02    0.00   NaN         NaN
1  34.06    3.68    5.50    3.77   NaN         NaN
2  30.70    6.80   16.40    2.25   NaN         NaN
3  36.08    4.43   10.14    1.00   NaN         NaN
4  39.31    7.01   18.89    2.78   NaN         NaN

Dataset Shape:
(29531, 16)

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29531 entries, 0 to 29530
Data columns (total 16 columns):
#   Column      Non-Null Count  Dtype
---  -
0   City        29531 non-null  object
1   Date        29531 non-null  object
2   PM2.5       24933 non-null  float64
3   PM10       18391 non-null  float64
4   NO          25949 non-null  float64
5   NO2        25946 non-null  float64
6   NOx        25346 non-null  float64
7   NH3        19203 non-null  float64
8   CO         27472 non-null  float64
9   SO2        25677 non-null  float64
10  O3         25509 non-null  float64
11  Benzene    23908 non-null  float64
12  Toluene    21490 non-null  float64
13  Xylene     11422 non-null  float64
14  AQI        24850 non-null  float64
15  AQI_Bucket  24850 non-null  object
dtypes: float64(13), object(3)
memory usage: 2.6+ MB

```



AQI Analysis

Average AQI = 158.78155158985473

Maximum AQI = 2049.0

Minimum AQI = 13.0

Most Common AQI Category:

Moderate

CRISP-DM: Data understanding, preparation, modeling and evaluation completed.

SEMMA: Sample, Explore, Modify, Model and Assess completed.

KDD: Selection, Preprocessing, Transformation, Data Mining and Interpretation completed.

Air quality patterns and trends identified successfully.

Result: